

Program : Diploma in Printing Technology	
Course Code : 5101	Course Title: Packaging Techniques
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 45

Course Objectives:

- To provide concrete knowledge on fundamentals of packaging technology.
- To guide the student to the basic concepts of the packaging like its functions, importance, need, medium, variations of packaging techniques, machines, legal-security and marketing aspects, packaging tests, trends etc.
- To think and understand the packaging concept together with the printing aspects.
- To develop idea on the packaging trends and exciting possibilities of packaging in future.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Packaging medium		Materials for Printing & Packaging	3
Finishing operations		Print Finishing & Conversion Techniques	4

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the importance and concepts of packaging.	15	Understanding
CO2	Explain about various packaging materials and related finishing operations.	14	Understanding
CO3	Extend knowledge on various packaging techniques.	15	Understanding
CO4	Extend knowledge on various packaging machineries and trends in packaging.	14	Understanding
	Series Test	2	

CO-PO mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the importance and basic concepts of packaging.		
M1.01	Packaging: Introduction.	2	Understanding
M1.02	Functions of packaging	2	Understanding
M1.03	Types of packaging	2	Understanding
M1.04	Correlation of printing and packaging.	1	Understanding
M1.05	Package design concept.	2	Understanding
M1.06	Marking on packages.	2	Understanding
M1.07	Tests on packages.	4	Understanding
Contents: Packaging: Introduction- Definition, Importance of packaging. Functions of packaging - protection, utility, communication, containment. Types-Flexible packaging, Rigid packaging, Consumer packaging, Industrial packaging, Military packaging. Correlate the possibilities of printing with packaging. Factors influencing design of package. Markings on package - Handling marks, Routing marks, Information marks. Tests on package: Physical & Mechanical test- Drop test, Vibration test, Compression test, Inclined impact test, Rolling test, Drum test. Chemical & Climatic test - Rain test, Sand and Dust test, Salt Spray test, Fungus Resistance test, Cobb test.			
CO2	Explain about various packaging materials and packaging-oriented finishing operation.		
M2.01	Packaging materials.	3	Understanding
M2.02	Cushioning Materials	1	Understanding
M2.03	Adhesives.	2	Understanding
M2.04	Adhesive tapes.	2	Understanding
M2.05	Closures.	2	Understanding
M2.06	Finishing operations - Lamination, Coating, Varnishing, Sealing and Labeling.	4	Understanding
	Series Test I	1	

Contents:

Packaging materials: Wood, Board, Glass, Plastic, Metals, Glass, Corrugated board-Paper, Plastic and Fiber Board. Cushioning Materials - Functions, Properties. Adhesive: Functions, Properties, Types - Natural and Synthetic, Solvent Responsive Adhesive, Pressure Sensitive Adhesive, Heat sensitive adhesive, Chemically reactive adhesive. Adhesive tapes-fabric tapes, Paper tapes, Film tapes, Foil tapes, Foam tapes, Two faced tapes. Closure: Features and functions and types. Finishing operations: Definition, introduction, Lamination: Functions, Properties, Types - Wet lamination, Dry lamination, Hot melts lamination, Extrusion lamination. Coatings - Functions, Properties, Types - Primer coating, Protective coating, Heat sealable and Self adhesive coating. Varnishing: Features, importance and Types - Oil Varnish, Synthetic Varnishing, Spirit varnish, Cellulose varnish, Water varnish, Roller coating, Lacquers. Sealing: Definition, purpose, Methods- Heat sealing, Cold sealing. Labeling: definition, purpose, basic elements of labeling.

CO3	Extend knowledge on various packaging techniques.		
M3.01	Thermoforming.	1	Understanding
M3.03	Molding.	1	Understanding
M3.04	Thermoform packaging.	1	Understanding
M3.05	Stretch Wrapping & Strip packaging.	2	Understanding
M3.06	Active and Passive packaging	3	Understanding
M3.07	Aerosol packaging.	2	Understanding
M3.08	Aseptic packaging.	2	Understanding
M3.09	Intelligent packaging.	2	Understanding
M3.10	Advanced packaging techniques.	1	Understanding

Contents:

Thermoforming: Vacuum forming, Drape forming, Snap back forming, Plug assist forming, Pressure forming. Molding- Compression Molding, Transfer Molding, Injection blow molding, Extrusion blow molding. Thermoform packaging: Skin packaging, Shrink packaging, and Blister packaging. Stretch Wrapping, Strip packaging. Active packaging: Vacuum packaging, Gas packaging. Passive Atmosphere Modification- MAP, CAP. Aerosol packaging- Features, importance, Components and Filling methods. Aseptic packaging- Features, importance, Components and Filling methods. Intelligent Packaging. Advanced packaging techniques- overview.

CO4	Extend knowledge on various packaging machineries and trends in packaging.		
M4.01	Diverse aspects of various Packaging machines	3	Understanding
M4.02	Carton making.	2	Understanding
M4.03	Corrugated board Manufacturing.	3	Understanding
M4.04	Box manufacturing.	3	Understanding
M4.05	Packaging rules and regulations.	2	Understanding
M4.06	Innovative trends in packaging.	1	Understanding
	Series Test II	1	

Contents:

Component description, purpose and basic working principles of various Packaging machines. Carton making-Introduction, design terminology, carton making procedures. Corrugated board Manufacturing: Materials, components and process. Box manufacturing-box styles, box designs, joints, closing styles. Packaging rules and regulations. FDA, FPO, MFPO. Innovative trends in packaging.

Text / Reference:

T/R	Book Title/Author
T1	IIP- Packaging technology- Vol – I, II, III- IIP India .
R1	Frank Paine- Packaging design and performance.
R2	Jhonn Briston- Advance in plastic packaging technology.
R3	Kit L. Yam- Encyclopedia of Packaging Technology- A John Wiley & Sons, Inc., Publication.

Online resources:

Sl.No	Website Link
1	http://printwiki.org/Packing
2	https://www.google.co.in/books/edition/Packaging_Technology
3	https://www.packagingtechtoday.com/
4	https://iip-in.com/about_iip/about-iip.aspx
5	https://foodpackaging.foodtechconferences.org/